
A Median Perspective on Unlabeled Data for Out-of-Distribution Detection

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Abstract

1 Out-of-distribution (OOD) detection is indispensable for the robustness and reli-
2 ability of machine learning systems deployed in real-world applications. Recent
3 approaches leverage unlabeled data to strengthen OOD detection, yet exploiting
4 unlabeled in-the-wild data remains a hard problem: such data mixes in-distribution
5 (InD) and OOD samples, and the absence of a clean set of OOD examples obstructs
6 the training of an effective OOD classifier. We resolve this problem with Medix, a
7 novel framework that identifies outliers from unlabeled data through the median
8 operation. The median is the right tool here: it yields a stable estimate of the
9 central tendency and is provably robust to noise and outliers. From the outliers
10 Medix extracts, together with labeled InD data, we train a robust OOD classifier.
11 On the theory side, we derive error bounds that establish Medix attains a low error
12 rate. Our experiments confirm these guarantees: Medix consistently outperforms
13 existing methods across the board in open-world settings, validating our theoretical
14 analysis.

15 1 Introduction

16 Deploying machine learning models in real-world applications inevitably exposes them to safety and
17 reliability challenges, chief among them the presence of out-of-distribution (OOD) data. These OOD
18 samples, drawn from unknown categories, must not be predicted by the model. Yet neural networks
19 are inherently vulnerable and, in practice, lack any built-in mechanism to detect and appropriately
20 handle OOD inputs (Nguyen et al., 2015).

21 Identifying OOD samples at inference is critical and genuinely hard: models never see unknown
22 distributions during training and therefore cannot, on their own, distinguish OOD from in-distribution
23 (InD) data. Recent approaches (Katz-Samuels et al., 2022a; Du et al., 2024a) confront this challenge
24 by leveraging additional “in-the-wild” data to improve OOD detection. In particular, Katz-Samuels
25 et al. (2022a) uses unlabeled wild data to regularize model training while still classifying labeled
26 InD data. The appeal of such unlabeled wild data is clear: it is readily collected once a model is
27 deployed in its operating environment, and it lets the model capture the true distribution of OOD data
28 encountered at test time, yielding markedly more robust OOD detection.

29 Leveraging unlabeled wild data, however, is far from trivial because InD and OOD data are entangled
30 in a complex mixture. The absence of a distinct, clean set of OOD samples is precisely what makes
31 robust OOD detection difficult: the detector observes only data drawn from this mixed distribution,
32 with no indication of whether any given sample is InD or OOD. The problem remains underexplored,
33 leaving substantial room for advancement. Crucially, almost no prior work establishes a formal
34 theoretical foundation; to the best of our knowledge, Du et al. (2024a) is the only study to provide
35 one for the “in-the-wild” setting.

36 Meanwhile, recent studies have established the effectiveness of median-based approaches in data
 37 pruning (Acharya et al., 2024). Building on these developments, this paper sets out to answer the
 38 following question:

How can median-based methods leverage unlabeled wild data to facilitate OOD
 detection with theoretical guarantees?

39 We answer this question affirmatively by introducing a new median-centric perspective for OOD
 40 detection. Concretely, we propose a median-based optimization framework and develop Medix, an
 41 algorithm that identifies OOD samples from wild data with a low error rate. We back this algorithm
 42 with a theoretical foundation that guarantees minimal error, and we validate it experimentally,
 43 establishing the robustness and efficiency of our approach. Ours is among the very few works to
 44 provide such a theoretical foundation for the unlabeled “in-the-wild” setting. We prove that median-
 45 based filtering is robust for outlier detection in unlabeled mixtures by bounding the fraction of InD
 46 samples incorrectly flagged as outliers. This fraction is governed by two effects: the contamination
 47 effect, which quantifies the impact of OOD points and stays controlled whenever the OOD proportion
 48 is below 50%; and the concentration effect, which exploits the sub-Gaussian nature of InD gradients
 49 to bound deviations of InD points from their mean gradient. Together, these effects guarantee that,
 50 with high probability, the number of misclassified InD points remains small—so the method holds up
 51 even in the worst case.

52 We benchmark our approach against two categories of methods: (1) those trained solely on InD
 53 data, and (2) those trained with both InD data and an auxiliary unlabeled dataset. On CIFAR-100
 54 (Krizhevsky et al., 2009), Medix delivers a decisive improvement over the strong baseline KNN+ (Sun
 55 et al., 2022), outperforming it by an average of 40.98% in FPR95. Unlike approaches such as Outlier
 56 Exposure (Hendrycks et al., 2019), which depend on a clean, auxiliary unlabeled dataset—thereby
 57 imposing the strong distributional assumption that the auxiliary data is completely separable from the
 58 InD data—Medix achieves superior results without any such assumption, and is therefore far more
 59 flexible. Against WOODS (Katz-Samuels et al., 2022a), Medix further reduces the average FPR95
 60 by 1.32% on CIFAR-100 and 2.60% on CIFAR-10.

61 Our key contributions are as follows:

- 62 **C1)** We propose Medix, a median-centric greedy approach that filters outliers from unlabeled
 63 wild data and then trains an OOD detector on the identified outliers together with the InD
 64 samples. Its core novelty is the filtering stage.
- 65 **C2)** We establish theoretical guarantees for the robustness of median-based filtering in identi-
 66 fying both inliers and outliers within unlabeled mixtures. Specifically, we prove that the
 67 misclassification rates for both InD samples flagged as outliers and OOD samples retained as
 68 inliers are tightly controlled by two effects: the contamination effect, which stays bounded
 69 whenever the proportion of OOD samples is below 50%, and the concentration effect, which
 70 guarantees stability in the gradient behavior of InD samples.
- 71 **C3)** We conduct an extensive evaluation of Medix across eleven InD-OOD pairs, comparing
 72 it against 20 competitive baselines. Medix outperforms every one of them, delivering
 73 superior performance across the board. Our experiments further show that Medix outlier
 74 extraction achieves a low error rate (e.g., only 12.5%; see Figure 2), directly corroborating
 75 our theoretical findings.

76 2 Preliminaries and Problem Setup

77 In this section, we first provide an overview of the OOD detection problem, and then formally define
 78 the data setup, model architecture, loss functions, and the learning goal.

79 **Labeled In-Distribution Data.** Consider the input space $\mathcal{X}$ and the label space $\mathcal{Y} = \{1, \dots, K\}$,
 80 which together define the structure of InD data. A labeled dataset $\mathcal{S}_{\text{in}} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$
 81 is generated by sampling n pairs independently and identically distributed (i.i.d.) from $\mathbb{P}_{XY}$, an
 82 unknown joint probability distribution over $\mathcal{X} \times \mathcal{Y}$. The marginal distribution of $\mathbb{P}_{XY}$ on $\mathcal{X}$ is denoted
 83 as $\mathbb{P}_{\text{in}}$, representing the underlying distribution of InD inputs. We use $\mathcal{S}_{\text{in}}$ to train an InD model.

84 **Out-of-distribution detection.** We address a practical scenario where the model is trained using
 85 labeled InD data but is later deployed in environments that may contain OOD inputs from classes not

86 represented in the training data, i.e., for some label $y \notin \mathcal{Y}$. The model is expected to abstain from
 87 making predictions for such OOD inputs. At inference time, the primary objective is to determine
 88 whether a given input belongs to the InD distribution or arises from an OOD source.

89 **Unlabeled wild data.** A primary obstacle in OOD detection is the scarcity of labeled OOD samples.
 90 Because the potential sample space of OOD data is vast, collecting labeled examples is both costly
 91 and impractical. We overcome this by introducing unlabeled wild data, $\mathcal{S}_{\text{wild}} = \{\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_m\}$,
 92 into our learning framework to faithfully mimic real-world scenarios, following Katz-Samuels et al.
 93 (2022a). Wild data blends InD and OOD samples and is readily collected during the deployment
 94 phase of a pre-trained model on $\mathcal{S}_{\text{in}}$. Similar to Du et al. (2024a); Katz-Samuels et al. (2022a), we
 95 adopt the Huber contamination model (Huber, 1964) to characterize the marginal distribution of the
 96 wild data

$$\mathbb{P}_{\text{wild}} := (1 - \pi)\mathbb{P}_{\text{in}} + \pi\mathbb{P}_{\text{out}}, \quad \pi \in (0, 1], \quad (1)$$

97 where π denotes the contamination proportion and $\mathbb{P}_{\text{out}}$ captures the OOD distribution over $\mathcal{X}$. The
 98 case $\pi = 0$ corresponds to the absence of OOD samples and renders the problem trivial.

99 **Models and Loss Functions.** Let $f_\phi : \mathcal{X} \rightarrow \mathbb{R}^K$ represent the InD classifier parameterized by
 100 $\phi \in \Phi$, where Φ denotes the parameter space for this classifier. The output of f_ϕ corresponds to a
 101 soft probability distribution over the $K = |\mathcal{Y}|$ InD classes. The loss function for the labeled InD data
 102 is defined as $\ell : \mathbb{R}^K \times \mathcal{Y} \rightarrow \mathbb{R}$. For OOD detection, we introduce a separate classifier $g_\theta : \mathcal{X} \rightarrow \mathbb{R}$,
 103 parameterized by $\theta \in \Theta$, with Θ as the parameter space. The binary loss function associated with
 104 g_θ is denoted as $\ell_b(g_\theta(\mathbf{x}), y_b)$, where $y_b \in \mathcal{Y}_b := \{y_+, y_-\}$. Here, $y_+ > 0$ represents the InD class,
 105 while $y_- < 0$ corresponds to the OOD class.

106 **Learning objective.** Our learning framework is designed to simultaneously train the OOD detector
 107 g_θ and the multi-class classifier f_ϕ , leveraging both the InD data $\mathbb{P}_{\text{in}}$ and the wild data $\mathbb{P}_{\text{wild}}$. During
 108 testing, we evaluate the performance using the following metrics:

$$\begin{aligned} \downarrow \text{FPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{out}}^{\text{test}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{TPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{in}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{Acc}(f_\phi) &= \mathbb{E}_{(\mathbf{x}, y) \sim \mathbb{P}_{\mathcal{X} \times \mathcal{Y}}} [\mathbb{I}\{f_\phi(\mathbf{x}) = y\}], \end{aligned}$$

109 where $\mathbb{I}\{\cdot\}$ denotes the indicator function, and $\mathbb{P}_{\text{out}}^{\text{test}}$ represents the OOD test data distribution.

110 3 Method: Median-Centric Framework for OOD Detection

111 In this section, we present a novel learning paradigm, **Medix**, that harnesses the power of unlabeled
 112 wild data for OOD detection. Our framework overcomes the fundamental limitation of conventional
 113 approaches that rely exclusively on InD data, and it is purpose-built for open-world environments
 114 where models routinely face previously unseen inputs. Medix comprises two integral stages: **1)**
 115 **Outlier Extraction:** a filtering process that isolates candidate OOD samples from the unlabeled
 116 wild data (Section 3.1), and **2) Detector Training:** training a binary OOD detector on both InD data
 117 and the outlier candidates identified in the previous step (Section 3.2). For stage 2, we follow the
 118 protocol of Du et al. (2024a). As the subsequent sections demonstrate, this two-stage methodology
 119 both extracts OOD data effectively from the unlabelled wild data and establishes a robust foundation
 120 for deploying machine learning models in dynamic, open-world scenarios.

121 3.1 Extracting Candidate Outliers from the Wild Data

122 To isolate potential outliers from the wild mixture $\mathcal{S}_{\text{wild}}$, our framework adopts an optimization-
 123 based approach that exploits the gradients of the model parameters. These gradients come from a
 124 classification model, f_ϕ , trained solely on the InD dataset $\mathcal{S}_{\text{in}}$. We formalize the methodology below.

125 **Reference gradient estimation from InD data.** The first step in our proposed framework is to
 126 estimate a reference gradient using the InD dataset $\mathcal{S}_{\text{in}}$. This is achieved by training a classifier f_ϕ on
 127 $\mathcal{S}_{\text{in}}$ through empirical risk minimization (ERM) as follows

$$\phi_{\mathcal{S}_{\text{in}}} \in \arg \min_{\phi \in \Phi} \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi), \quad \text{where } \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi) = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \ell(f_\phi(\mathbf{x}_i), y_i), \quad (2)$$

where $\phi_{S_{\text{in}}}$ denotes the learned parameters. Once the classifier has been trained, we compute the mean gradient $\bar{\nabla}_{\text{in}}$ as the average of the gradients of the loss function with respect to the model parameters over the InD data:

$$\bar{\nabla}_{\text{in}} = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in S_{\text{in}}} \nabla \ell(f_{\phi_{S_{\text{in}}}}(\mathbf{x}_i), y_i). \quad (3)$$

In our approach, $\bar{\nabla}_{\text{in}}$ serves as the reference gradient, enabling the quantification of deviations for other data points relative to this reference.

Motivation. We hypothesize that increasing the number of OOD samples in the wild dataset, S_{wild} , drives a greater deviation from the average InD gradient, $\bar{\nabla}_{\text{in}}$. To test this hypothesis directly, we design an initial experiment using CIFAR-10 (Krizhevsky et al., 2009) as the InD dataset and SVHN (Netzer et al., 2011) as the OOD dataset. Here S_{wild} consists of 10,000 samples drawn from CIFAR-10, kept disjoint from the training set used to train the model $\phi_{S_{\text{in}}}$, which we use to compute $\bar{\nabla}_{\text{in}}$. We incrementally add SVHN OOD samples to S_{wild} and track the L_2 -norm deviation between $\bar{\nabla}_{\text{in}}$ and the element-wise median (EWM) of the gradients of the wild dataset:

$$\left\| \bar{\nabla}_{\text{in}} - \text{EWM} \left(\left\{ \nabla \ell \left(f_{\phi_{S_{\text{in}}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in S_{\text{wild}}} \right) \right\|.$$

Our results, shown in Figure 1, reveal a clear, monotonic increase in the L_2 -norm deviation, confirming our hypothesis. This observation directly motivates the method we introduce in the next section. Indeed, the stopping criterion for our algorithm follows from this monotonically increasing behavior: we terminate once the L_2 -norm deviation between consecutive iterations drops below a threshold ϵ , as we detail in the following section.

Filtering potential outliers from unlabeled wild data. Motivated by the results in Figure 1, we formulate the following optimization problem to identify the outlier subset S_{out}^* in S_{wild} :

$$S_{\text{in}}^* = \arg \min_{S \subseteq S_{\text{wild}}} \left\| \bar{\nabla}_{\text{in}} - \text{EWM}(G_S) \right\|, \text{ where } G_S = \left\{ \nabla \ell \left(f_{\phi_{S_{\text{in}}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in S}. \quad (4)$$

Here $\text{EWM}(\cdot)$ denotes the element-wise median function, and $\hat{y}_{\tilde{\mathbf{x}}_i}$ represents the predicted label for a wild sample $\tilde{\mathbf{x}}_i$. For notational simplicity, we define $G_S = \{\nabla \ell(\tilde{\mathbf{x}}_i)\}_{i \in S}$. This optimization problem identifies the subset S in S_{wild} that minimizes the distance between the EWM of the gradients and the average gradient $\bar{\nabla}_{\text{in}}$. As Figure 1 makes clear, such a subset, denoted S_{in}^* , represents the InD data in S_{wild} , so that its complement $S_{\text{out}}^* = S_{\text{wild}} \setminus S_{\text{in}}^*$ captures the OOD data in S_{wild} .

Solving the optimization problem in (4) exactly is computationally prohibitive as the size of S_{wild} grows. We therefore introduce an efficient greedy approximation based on a leave-one-out approach, outlined in Algorithm 1. The algorithm implements an iterative procedure for outlier detection from a wild dataset, exploiting deviations in gradient information relative to the InD dataset.

The algorithm begins by computing the EWM of the wild data gradients at each iteration, which serves as a reference for comparing against the average gradient of the InD data $\bar{\nabla}_{\text{in}}$. We denote by d_t , the L_2 distance between the average InD gradient $\bar{\nabla}_{\text{in}}$ and the EWM gradients of the data left in the wild set, represented by S , i.e., $d_t = \|\text{EWM}(G_S) - \bar{\nabla}_{\text{in}}\|$. The algorithm then iteratively identifies samples in S that incur the most significant drop in the L_2 distance with d_t when removed from S as OOD. Specifically, having data samples S left, we remove each sample $i \in S$ and compute the EWM as $\text{EWM}(G_{S \setminus \{i\}})$ and the distance to $\bar{\nabla}_{\text{in}}$ as $\|\text{EWM}(G_{S \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$. We then find the drop in distance $\delta_i = d_t - \|\text{EWM}(G_{S \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$ and find the k samples with the largest drop and identify them as OOD. The algorithm repeats until there is no significant drop in δ_i or a maximum number of iterations is reached. The convergence criterion is the change in the L_2 distance between two iterations, which must fall below a predefined ϵ threshold. This criterion follows directly from the monotonically increasing trend we observed in our preliminary experiment (Figure 1): once a

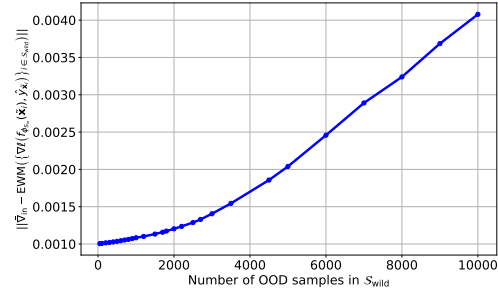


Figure 1: Distance deviation grows monotonically as we increase OOD samples in S_{wild} .

substantial number of OOD samples have been removed, the L_2 distance decreases to a small value, precisely signaling when the algorithm should halt to avoid erroneously flagging InD samples as outliers and thereby preventing any degradation in performance.

Algorithm 1 Iterative Outlier Detection via **Medix**

Require: $\bar{\nabla}_{\text{in}}, G_i = \nabla \ell(f_{\phi_{S_{\text{in}}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i})$, maximum iterations T , hyperparameters ϵ, k

- 1: Initialize $\mathcal{S} \leftarrow \mathcal{S}_{\text{wild}}$ (wild set), $\mathcal{O} \leftarrow \emptyset$ (outliers), $t \leftarrow 0$ (iteration), $d_t \leftarrow 0$, $\delta_{\text{max}} \leftarrow \infty$ (deviations), $\mathcal{I}_k \leftarrow \emptyset$
- 2: **while** $t \leq T$ or $|\delta_{\text{max}}| > \epsilon$ **do**
- 3: $\mathcal{O} \leftarrow \mathcal{O} \cup \{\tilde{\mathbf{x}}_i : i \in \mathcal{I}_k\}$ ▷ Add outliers to set $\mathcal{O}$
- 4: $d_t \leftarrow \|\text{EWM}(G_{\mathcal{S}}) - \bar{\nabla}_{\text{in}}\|$ ▷ Compute L2 deviation
- 5: **for each** $i \in \mathcal{S}$ **do**
- 6: $\delta_i \leftarrow d_t - \|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$
- 7: **end for**
- 8: $\mathcal{I}_k \leftarrow \text{indices}(\text{top-}k(\{\delta_i\}_{i \in \mathcal{S}}))$ ▷ Select top- k indices
- 9: $\mathcal{S} \leftarrow \mathcal{S} \setminus \mathcal{I}_k$ ▷ Remove outliers from $\mathcal{S}$
- 10: $\delta_{\text{max}} \leftarrow \max_{i \in \mathcal{S}} \{\delta_i\}$, $t \leftarrow t + 1$
- 11: **end while**
- 12: **return** $\hat{\mathcal{S}}_{\text{out}} = \mathcal{O}$ ▷ Return detected outliers

3.2 Training the OOD Detector with Candidate Outliers

Having identified the candidate outlier set $\hat{\mathcal{S}}_{\text{out}}$ from the wild data, we train the OOD detector g_{θ} to maximize the distinction between InD and candidate outlier data, following the protocol in Du et al. (2024a). The objective explicitly enforces separability at the decision boundary (thresholded at 0), assigning positive outputs to labeled InD samples $\mathbf{x} \in \mathcal{S}_{\text{in}}$ and negative outputs to candidate outliers $\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}$. The loss function is defined as:

$$\begin{aligned} \mathcal{L}_{\mathcal{S}_{\text{in}}, \hat{\mathcal{S}}_{\text{out}}}(g_{\theta}) &= \mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_{\theta}) + \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^-(g_{\theta}), \\ \mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_{\theta}) &= \mathbb{E}_{\mathbf{x} \in \mathcal{S}_{\text{in}}} \mathbb{I}\{g_{\theta}(\mathbf{x}) \leq 0\}, \\ \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^-(g_{\theta}) &= \mathbb{E}_{\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}} \mathbb{I}\{g_{\theta}(\tilde{\mathbf{x}}) > 0\}. \end{aligned} \quad (5)$$

To handle the non-differentiable 0/1 loss, we employ a binary loss based on a differentiable sigmoid function as a smooth surrogate, which makes optimization tractable while remaining faithful to the original objective. The framework also incorporates the InD risk defined in Equation 2, safeguarding predictive accuracy on InD samples. This unified optimization substantially strengthens the generalization of g_{θ} and enables robust detection of OOD samples drawn from $\mathbb{P}_{\text{out}}$, as our results confirm.

4 Theoretical Analysis

We now present the theoretical guarantees underlying Medix’s filtering stage. The following theorems establish provable upper bounds on the misclassification rates for both InD and OOD points, and together they demonstrate the two-sided robustness of our EWM filtering. Detailed proofs appear in Appendix C.

Theorem 4.1 (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$, and fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the EWM filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

Theorem 4.1 shows that the inlier misclassification rate is governed by exactly two effects: a *concentration term*, which vanishes as the sample size grows and captures statistical fluctuations in

the InD gradients, and a *contamination term*, which quantifies the influence of OOD points on the EWM. Critically, the bound stays controlled for any contamination ratio $\pi < 0.5$, establishing the robustness of median-based filtering to moderate OOD presence. We now address the complementary question: *how many true outliers escape detection and are mistakenly retained?* The next theorem answers it by bounding the OOD misclassification rate.

Theorem 4.2 (Outlier Misclassification Bound). *Assume that the gradients of OOD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ_{out}^2 . Suppose the mean OOD gradient μ_{out} satisfies a separation condition:*

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d},$$

for some $\Delta > 0$. Then, for any tolerance $\epsilon \in (0, \Delta)$ and confidence level $\delta \in (0, 1)$, with probability at least $1 - \delta$, the outlier misclassification rate satisfies:

$$\text{ERR}_{\text{out}} \leq \underbrace{2d \exp\left(-\frac{(\Delta - \epsilon)^2}{2\sigma_{\text{out}}^2}\right)}_{\text{Separation term}} + \underbrace{\sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}}_{\text{Concentration term}} + \underbrace{\frac{1 - \pi}{2\pi}}_{\text{Contamination term}}.$$

Theorem 4.2 complements the previous result by bounding the fraction of OOD points mistakenly retained as InD. Together, Theorems 4.1 and 4.2 deliver a complete two-sided guarantee for the robustness of Medix filtering. The inlier and outlier misclassification rates are governed by the balance of three effects:

Contamination Effect. In Theorem 4.1, even when the wild dataset contains a significant fraction π of OOD points, the median-based criterion remains stable for all $\pi < 0.5$. This is captured by the $\pi/[2(1 - \pi)]$ term of the bound, which grows with π and encodes the risk that OOD gradients skew the EWM. Symmetrically, Theorem 4.2 carries a penalty $\frac{1 - \pi}{2\pi}$ that quantifies the difficulty of isolating OOD points when they are underrepresented.

Concentration Effect. Both bounds exploit the sub-Gaussian nature of the gradient coordinates. For InD data, this guarantees that most gradients concentrate near $\bar{\nabla}_{\text{in}}$, keeping the median stable even in finite samples. For OOD points, concentration around μ_{out} lets us pin down the risk that a misaligned OOD sample slips past the filter. These concentration effects decay as $1/\sqrt{m}$, where m is the number of samples from each class.

Separation Effect. Unique to Theorem 4.2, this effect quantifies exactly how far the OOD mean gradient must lie from the InD mean gradient to reliably reject OOD samples. The exponential term $\exp(-(\Delta - \epsilon)^2/2\sigma_{\text{out}}^2)$ captures the trade-off precisely: the more separated the distributions, the less likely OOD gradients are to fool the filter.

Theorems 4.1 and 4.2 jointly prove that, with high probability, median filtering achieves robust separation of InD and OOD samples in unlabeled data mixtures. The fraction of InD samples misclassified as outliers is bounded by contamination and concentration effects, while the fraction of OOD points incorrectly retained is governed by an exponential separation term, a concentration bound, and a reverse contamination effect. These results give a rigorous theoretical guarantee that Medix minimizes both types of error under mild assumptions.

Remark 4.3. *Sub-Gaussian Assumption for Gradient Coordinates.* The assumption that the gradients of InD samples are sub-Gaussian in each coordinate is what enables our concentration bounds and ensures robust statistical estimation, and our empirical evidence supports it directly. As Figure 4a shows, the histogram of InD gradient values is bell-shaped and tightly concentrated around the mean. The gradients therefore exhibit light tails—the defining, exponentially decaying tail behavior of sub-Gaussian random variables—so extreme gradient values are rare, exactly as the histogram reveals. Figure 4b reinforces this: it compares the empirical quantiles of the gradient distribution against the theoretical quantiles of a Gaussian in a Q-Q plot (Wilk and Gnanadesikan, 1968). The points align closely with the 45-degree reference line, showing that the empirical gradient distribution closely matches a Gaussian. Since Gaussian variables are a subset of sub-Gaussian variables, this confirms the sub-Gaussian assumption. We further provide a looser version of these bounds that **removes the sub-Gaussian assumption entirely and holds under merely bounded second moments**; see Theorem

233 *C.3 (Appendix C.3). Although the rates degrade, the core robustness guarantee of Medix continues to*
 234 *hold in this broader setting.*

235 5 Experiments

236 This section demonstrates the efficacy of Medix across a wide range of InD-OOD dataset pairs,
 237 benchmarking it against 20 widely-used baselines. All experiments run on hardware equipped with
 238 NVIDIA A100-SXM4-80GB GPUs. We will release the code needed to reproduce our results upon
 239 acceptance.

240 5.1 Models, Datasets, and Baselines

241 **Datasets.** We adopt exactly the experimental protocol of Katz-Samuels et al. (2022a), which
 242 introduced the problem of learning OOD detectors with wild data, ensuring a fair comparison to
 243 prior work. We use CIFAR-10 and CIFAR-100 as InD datasets ($\mathbb{P}_{\text{in}}$). For OOD testing, we use a
 244 suite of natural image datasets: PLACES365 (Zhou et al., 2017), SVHN, TEXTURES (Cimpoi et al.,
 245 2014), and LSUN-RESIZE & LSUN-C (Yu et al., 2015) as OOD datasets ($\mathbb{P}_{\text{out}}$). To simulate wild
 246 data ($\mathbb{P}_{\text{wild}}$), we combine a subset of InD data ($\mathbb{P}_{\text{in}}$) with the OOD data ($\mathbb{P}_{\text{out}}$) under a default mixing
 247 parameter $\pi = 0.5$. For instance, with PLACES365 as the OOD test set, we build a wild mixture
 248 by combining CIFAR with PLACES365 and test on PLACES365 as the OOD set. We repeat this
 249 procedure across all OOD datasets and baselines. We split the InD CIFAR dataset into two halves: the
 250 first 25,000 images train $\phi_{S_{\text{in}}}$, and the remaining images generate the wild mixture S_{wild} . For gradient
 251 computation we use the penultimate-layer weights, which are known to be particularly informative
 252 for OOD detection (Huang et al., 2021a).

253 **Evaluation metrics.** We evaluate using three standard metrics: (1) False Positive Rate (FPR95↓)
 254 of OOD samples when the True Positive Rate of InD samples is 95%, (2) Area Under the Receiver
 255 Operating Characteristic Curve (AUROC↑), and (3) InD Classification Accuracy (InD Acc↑).

256 **Baselines.** Our comparison encompasses a diverse set of competitive baselines, categorized based
 257 on whether they are trained using only InD data or both InD and wild data. For the methods trained
 258 exclusively on InD data ($\mathbb{P}_{\text{in}}$), we compare Medix against a variety of established OOD detection
 259 methods, including Maximum Softmax Probability (MSP) (Hendrycks and Gimpel, 2016), ODIN
 260 (Liang et al., 2018), Mahalanobis Distance (Lee et al., 2018a), Energy Score (Liu et al., 2020b), ReAct
 261 (Sun et al., 2021), DICE (Sun and Li, 2022), KNN Distance (Sun et al., 2022), and ASH (Djurisic
 262 et al., 2022); these methods use a model trained with softmax cross-entropy loss. Additionally, we also
 263 compare against methods based on contrastive loss, such as CSI (Tack et al., 2020a) and KNN+ (Sun
 264 et al., 2022), for a more comprehensive comparison. For methods that leverage both InD and wild
 265 data, we compare against Outlier Exposure (OE) (Hendrycks et al., 2019) and energy-regularization
 266 learning (Liu et al., 2020b), which regularize training by promoting lower confidence or higher energy
 267 on outlier data. We also compare against WOODS (Katz-Samuels et al., 2022a), which introduced
 268 the concept of wild unlabeled data and exploits it for OOD detection via constrained optimization.
 269 Finally, we include the most recent baselines, CONJ (Peng et al., 2024) and DRL (Zhang et al., 2024),
 270 for a thorough evaluation.

271 5.2 Experimental Setup

272 In line with WOODS (Katz-Samuels et al., 2022a), we employ a Wide ResNet architecture
 273 (Zagoruyko, 2016) with 40 layers and a width factor of 2 for the InD classifier $\phi_{S_{\text{in}}}$. It is trained
 274 using stochastic gradient descent with a momentum of 0.9, weight decay of 0.0005, and an initial
 275 learning rate of 0.1. Training is performed for 100 epochs with cosine learning rate decay, a batch
 276 size of 128, and a dropout rate of 0.3. Hyperparameters ϵ and k used in the proposed method Medix
 277 are selected from the sets $\{5e-5, 5e-4, 5e-3, 5e-2\}$ and $\{4k, 7k, 10k, 20k\}$, respectively, taking
 278 into account dataset sizes and with the objective of maximizing OOD performance. For the OOD
 279 classifier g_{θ} , we initialize it with the pre-trained InD classifier $\phi_{S_{\text{in}}}$ and add a linear layer that performs
 280 binary classification using the penultimate-layer features. The learning rate for this classifier is set to
 281 0.001, and fine-tuning is done for 100 epochs as outlined in Equation (5). We combine the binary

classification loss with the InD classification loss, assigning a weight of 10 to the binary classification component. All other training parameters remain the same as those used for training $\phi_{S_{in}}$.

Table 1: OOD detection performance of Medix versus baselines on CIFAR-10 as InD data. Medix attains the best OOD detection result on every dataset. Performance averaged over five runs; best results in **bold**.

Methods	OOD Datasets												InD Acc $\uparrow$
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average		
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	
Using $\mathbb{P}_{in}$ only													
MSP	48.49	91.89	59.48	88.20	30.80	95.65	52.15	91.37	59.28	88.50	50.04	91.12	94.84
ODIN	33.35	91.96	57.40	84.49	15.52	97.04	26.62	94.57	49.12	84.97	36.40	90.61	94.84
Mahalanobis	12.89	97.62	68.57	84.61	39.22	94.15	42.62	93.23	15.00	97.33	35.66	93.34	94.84
Energy	35.59	90.96	40.14	89.89	8.26	98.35	27.58	94.24	52.79	85.22	32.87	91.73	94.84
KNN	24.53	95.96	25.29	95.69	25.55	95.26	27.57	94.71	50.90	89.14	30.77	94.15	94.84
ReAct	40.76	89.57	41.44	90.44	14.38	97.21	33.63	93.58	53.63	86.59	36.77	91.48	94.84
DICE	35.44	89.65	46.83	86.69	6.32	98.68	28.93	93.56	53.62	82.20	34.23	90.16	94.84
ASH	6.51	98.65	48.45	88.34	0.90	99.73	4.96	98.92	24.34	95.09	17.03	96.15	94.84
CSI	17.30	97.40	34.95	93.64	1.95	99.55	12.15	98.01	20.45	95.93	17.36	96.91	94.17
KNN+	2.99	99.41	24.69	94.84	2.95	99.39	11.22	97.98	9.65	98.37	10.30	97.99	93.19
Using $\mathbb{P}_{in}$ and $\mathbb{P}_{wild}$													
OE	1.13	99.53	19.48	94.88	1.91	98.16	0.54	98.84	7.75	98.56	6.16	97.99	94.12
Energy (w/ OE)	5.24	98.72	14.66	96.18	2.35	99.30	4.85	98.62	10.51	97.10	7.52	97.98	94.24
WOODS	0.17	99.91	10.19	98.05	0.31	99.14	0.11	99.38	6.21	98.13	3.40	98.92	94.74
Medix	0.06	99.98	2.98	99.10	0.01	99.98	0.01	99.98	0.96	99.66	0.80	99.74	93.58
(Ours)	± 0.01	± 0.01	± 0.29	± 0.09	± 0.01	± 0.00	± 0.01	± 0.01	± 0.13	± 0.06	± 0.09	± 0.03	± 0.64

Table 2: OOD detection performance of Medix versus baselines on CIFAR-100 as InD data. Medix attains the best OOD detection result on every dataset. Performance averaged over five runs; best results in **bold**.

Methods	OOD Datasets												InD ACC $\uparrow$
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average		
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	
Using $\mathbb{P}_{in}$ only													
MSP	84.59	71.44	82.84	73.78	66.54	83.79	82.42	75.38	83.29	73.34	79.94	75.55	75.96
ODIN	84.66	67.26	87.88	71.63	55.55	87.73	71.96	81.82	79.27	73.45	75.86	76.38	75.96
Mahalanobis	57.52	86.01	88.83	67.87	91.18	69.69	21.23	96.00	39.39	90.57	59.63	82.03	75.96
Energy	85.82	73.99	80.56	75.44	35.32	93.53	79.47	79.23	79.41	76.28	72.12	79.69	75.96
KNN	66.38	83.76	79.17	71.91	70.96	83.71	77.83	78.85	88.00	67.19	76.47	77.08	75.96
ReAct	74.33	88.04	81.33	74.32	39.30	91.19	79.86	73.69	67.38	82.80	68.44	82.01	75.96
DICE	88.35	72.58	81.61	75.07	26.77	94.74	80.21	78.50	76.29	76.07	70.65	79.39	75.96
ASH	21.36	94.28	68.37	71.22	15.27	95.65	68.18	85.42	40.87	92.29	42.81	87.77	75.96
CSI	64.70	84.97	82.25	73.63	38.10	92.52	91.55	63.42	74.70	92.66	70.26	81.44	69.90
KNN+	32.21	93.74	68.30	75.31	40.37	86.13	44.86	88.88	46.26	87.40	46.40	86.29	73.78
Using $\mathbb{P}_{in}$ and $\mathbb{P}_{wild}$													
OE	2.86	99.05	40.21	88.75	4.13	99.05	1.25	99.38	22.86	94.63	14.26	96.17	73.38
Energy (w/ OE)	2.71	99.34	34.82	90.05	3.27	99.18	2.54	99.23	30.16	94.76	14.70	96.51	72.76
WOODS	0.17	99.80	21.87	93.73	0.48	99.61	1.24	99.54	9.95	95.97	6.74	97.73	73.91
Medix	0.16	99.96	15.99	95.23	0.13	99.98	0.83	99.83	8.02	97.79	5.42	98.96	73.33
(Ours)	± 0.02	± 0.00	± 0.66	± 0.14	± 0.06	± 0.02	± 0.36	± 0.06	± 0.75	± 0.30	± 0.37	± 0.10	± 0.83

5.3 Results

We present our main results on CIFAR-100 in Table 2, where Medix decisively outperforms all OOD detection baselines. Two observations stand out. (1) Methods trained on both InD and wild data substantially outperform those trained exclusively on InD data: Medix cuts the FPR95 by 52.31% on PLACES365 and 38.24% on TEXTURES relative to KNN+, confirming the value of incorporating in-the-wild data for model regularization. (2) Medix also surpasses every competitive method that uses wild data ($\mathbb{P}_{wild}$): on CIFAR-100 it achieves an average FPR95 of 5.42%, a 1.32% improvement over WOODS, while maintaining a competitive InD accuracy of 73.33%. This small accuracy gap is fully explained by our using only 25,000 labeled InD samples, whereas the InD-only baselines use the full CIFAR-100 training set of 50,000 samples. Table 1 reports the CIFAR-10 results, where Medix again surpasses every baseline. It outperforms WOODS by 7.21% on PLACES365 and 5.25% on TEXTURES in FPR95, underscoring its effectiveness at detecting OOD samples.

A representative visual example of Medix. We further examine how Algorithm 1 (Outlier Extraction) extracts OOD samples from wild data $\mathcal{S}_{wild}$. To visualize this, we design a deliberately simple experiment on 2-dimensional synthetic data for clarity. We generate the InD data by sampling from three multivariate Gaussian distributions, one per class, with mean vectors $[-2, 0]$, $[2, 0]$, and $[0, 2\sqrt{3}]$. The covariance matrix for all three classes is fixed at $0.25 \cdot I$. For the OOD data, we use a multivariate Gaussian $\mathcal{N}([20, 2\sqrt{3}], 0.25 \cdot I)$. Figure 2 shows that Medix cleanly identifies the outliers, the vast majority of which align with the ground truth: it correctly flags 87.5% of the actual OOD samples, confirming its robustness at outlier extraction.

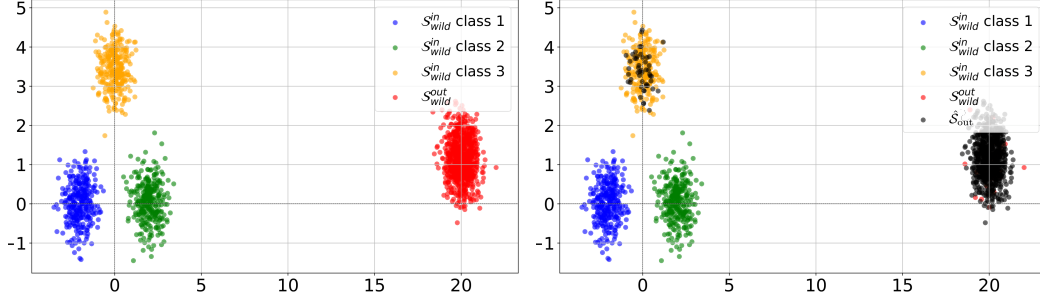


Figure 2: Medix applied to unlabeled wild data. (a) Setup of the InD data $\mathcal{S}_{\text{wild}}^{\text{in}}$ and OOD data $\mathcal{S}_{\text{wild}}^{\text{out}}$ in the wild, with inliers sampled from three multivariate Gaussian distributions. (b) Outliers $\hat{\mathcal{S}}_{\text{out}}^{\text{out}}$ filtered by Medix (in black): the error rate of $\hat{\mathcal{S}}_{\text{out}}^{\text{out}}$ containing InD data $\mathcal{S}_{\text{wild}}^{\text{in}}$ is a mere 12.5%.

Additional studies and insights. Due to space constraints, we defer additional experiments and insights to Appendix A: (1) ablation studies on hyperparameter selection and sensitivity analysis (Appendix A.2), which establish Medix’s strong robustness to hyperparameters; (2) a comparison between EWM and geometric median (Appendix A.1), which shows EWM is more sensitive to distributional shifts and hence the more effective and reliable filtering choice for our method; (3) a comprehensive comparison with methods employing competitive contrasting learning objectives (Appendix A.3), where Medix outperforms even these competitive baselines by a significant margin; (4) an evaluation on large-scale data under the complex unseen OOD setting, where $\mathbb{P}_{\text{out}}^{\text{test}} \neq \mathbb{P}_{\text{out}}$ (Appendix A.4), where Medix again outperforms baselines by a significant margin; (5) an evaluation of the computation and memory efficiency of Medix (Appendix A.6); (6) an evaluation of the impact of pseudo-label quality (Appendix A.5), which shows our method is resilient to noisy or low-confidence labels; and (4) a comparison with semi-supervised open-set recognition methods (Appendix A.7).

6 Related Work

In recent years, there has been a growing interest in OOD detection (Fort et al., 2021; Yang et al., 2024; Fang et al., 2022; Zhu et al., 2022; Yang et al., 2022; Wang et al., 2022c; Galil et al., 2023; Djurisić et al., 2023; Tao et al., 2023; Zheng et al., 2023; Wang et al., 2022b, 2023b; Uppaal et al., 2023; Zhu et al., 2023; Bai et al., 2023; Ming and Li, 2024; Zhang et al., 2023; Ghosal et al., 2024; Abbas et al., 2025). One approach to detect OOD data uses scoring functions to assess data distribution, including distance-based methods (Lee et al., 2018b; Tack et al., 2020b; Ren et al., 2021; Schwag et al., 2021; Sun et al., 2022; Du et al., 2022a; Ming et al., 2023; Ren et al., 2022), gradient-based score (Huang et al., 2021b), energy-based score (Liu et al., 2020c; Wang et al., 2021; Wu et al., 2023), confidence-based approaches (Bendale and Boult, 2016; Hendrycks and Gimpel, 2017; Liang et al., 2018), and Bayesian methods (Gal and Ghahramani, 2016; Lakshminarayanan et al., 2017; Maddox et al., 2019; Malinin and Gales, 2019; Wen et al., 2020; Kristiadi et al., 2020).

Another approach to OOD detection involves using regularization techniques during the training phase (Malinin and Gales, 2018; Geifman and El-Yaniv, 2019; Hein et al., 2019; Meinke and Hein, 2020; Jeong and Kim, 2020; Liu et al., 2020a; Van Amersfoort et al., 2020; Yang et al., 2021; Wei et al., 2022; Du et al., 2022b, 2023; Wang et al., 2023a). For example, regularization techniques can be applied to the model to either reduce its confidence (Lee et al., 2017; Hendrycks et al., 2019) or increase its energy (Liu et al., 2020c; Du et al., 2022c; Ming et al., 2022) on the OOD data. Most of these regularization methods assume the availability of an auxiliary OOD dataset.

Several studies (Zhou et al., 2021; Katz-Samuels et al., 2022b; He et al., 2023) relax the labeled-data-only assumption by incorporating unlabeled wild data (Katz-Samuels et al., 2022a; Geng et al., 2025), but none propose a clear mechanism for outlier detection. Du et al. (2024a,b) do introduce an explicit outlier filtering method, yet their thresholding technique differs fundamentally from ours: we detect outliers through a new median-centric approach. Moreover, Katz-Samuels et al. (2022a); Du et al. (2024a) assume batch-level mixing, in which each batch contains a fixed ratio of InD and OOD

samples. With large outsourced datasets this structured mixing simply does not exist—data is mixed randomly across the dataset. Our method removes this restriction, enabling dataset-level mixing without any reliance on batch-level structure. Many studies also leverage positive-unlabeled learning, which trains classifiers from positive and/or unlabeled data (Letouzey et al., 2000; Hsieh et al., 2015; Plessis et al., 2015; Niu et al., 2016; Gong et al., 2018; Chapel et al., 2020; Garg et al., 2021; Xu and Denil, 2021; Garg et al., 2022; Zhao et al., 2022; Acharya et al., 2022). The key distinction is that these methods focus solely on separating $\mathbb{P}_{\text{out}}$ from $\mathbb{P}_{\text{in}}$ and never train an OOD classifier. In contrast, we propose a median-centric method that identifies outliers in unlabeled data with provably low error rates.

7 Conclusions

We introduced Medix, a novel median-centric framework for OOD detection that leverages unlabeled in-the-wild data. Exploiting the inherent robustness of the median operation, Medix filters outliers from mixed unlabeled data and thereby enables the training of a reliable OOD detector. Our theoretical analysis established provable bounds on the inlier misclassification rate, proving that Medix stays robust even under significant OOD contamination (up to 50%), with errors controlled by sub-Gaussian concentration and contamination effects. We further provided complementary theoretical limits on the OOD misclassification rate, guaranteeing accurate isolation of OOD samples under clear separation conditions. Empirical validation across diverse benchmarks confirmed Medix’s superiority over 20 baselines: it reduced the average FPR95 by 40.98% relative to strong baselines such as KNN+, outperformed state-of-the-art methods including WOODS and DRL, and achieved an outlier extraction error rate as low as 12.5%.

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A Additional Studies

A.1 Element-wise Median vs Geometric Median

A natural question is why we chose the element-wise median over the geometric median (Acharya et al., 2024) for outlier detection in Medix. To answer it, we conducted a controlled experiment using CIFAR-10 as the InD dataset and SVHN as the OOD dataset. Here $\mathcal{S}_{\text{wild}}$ consists of 5k samples drawn from CIFAR-10, kept disjoint from the training set used to train the model $\phi_{\mathcal{S}_{\text{in}}}$, which we use to compute $\bar{\nabla}_{\text{in}}$. We incrementally add SVHN OOD samples to $\mathcal{S}_{\text{wild}}$ and track both the L_2 -norm deviation between $\bar{\nabla}_{\text{in}}$ and the element-wise median of the wild gradients and the proportion of OOD samples removed. The results in Figure 3 are unambiguous: for the same number of OOD samples in the wild dataset, the element-wise median identifies a substantially higher proportion of OOD samples as outliers than the geometric median. The element-wise median is therefore more sensitive to distributional shifts, making it the more effective and reliable filtering choice for our method.

Table 3: Medix stays robust across hyperparameters ϵ and k , outperforming all baselines.

Method	FPR95↓	ϵ	k
DICE	88.35	—	—
ASH	21.36	—	—
CSI	64.70	—	—
KNN+	32.21	—	—
OE	2.86	—	—
Energy	2.71	—	—
Medix	0.16	0.005	20000
Medix	0.20	0.0005	20000
Medix	0.68	0.005	10000

A.2 Hyperparameter Selection and Sensitivity Analysis

We conducted an ablation study to assess the sensitivity of Medix to the hyperparameters ϵ and k , using CIFAR-100 as the InD dataset and SVHN as the OOD dataset. As Table 3 shows, Medix is strongly robust to variations in ϵ and k . It achieves a remarkably low FPR95 of 0.16 at $\epsilon = 0.005$ and $k = 20000$. Even when we vary these values—reducing ϵ to 0.0005 or halving k to 10000—performance remains strong (FPR95 of 0.20 and 0.68, respectively), degrading gracefully rather than collapsing. Thus, while optimal tuning helps, Medix retains its effectiveness and continues to surpass the baselines across a wide range of hyperparameter choices. This insensitivity to exact tuning makes Medix a dependable choice for real-world deployments, where exhaustive tuning is rarely feasible.

A.3 Comprehensive comparison with recent competitive methods

We extend our comparisons to competitive baselines built on diverse contrasting learning objectives, including SSD+ (Sehwag et al., 2021), ProxyAnchor (Kim et al., 2020), and CIDER Ming et al. (2023). Although these methods rely on contrasting learning objectives that we do not use, we include them for a comprehensive comparison. We further include the most recent methods—CONJ (Peng et al., 2024), DRL (Zhang et al., 2024), Vim (Wang et al., 2022a), and VOS (Du et al., 2022c)—for a thorough evaluation. For this experiment, we use CIFAR-100 as InD dataset and train the ResNet-34 model¹ following the setup in Ming et al. (2023); Zhang et al. (2024). The model is trained

¹Since ResNet-34 is the standard model across these works, we adopt the same model to ensure consistency and facilitate a fair comparison.

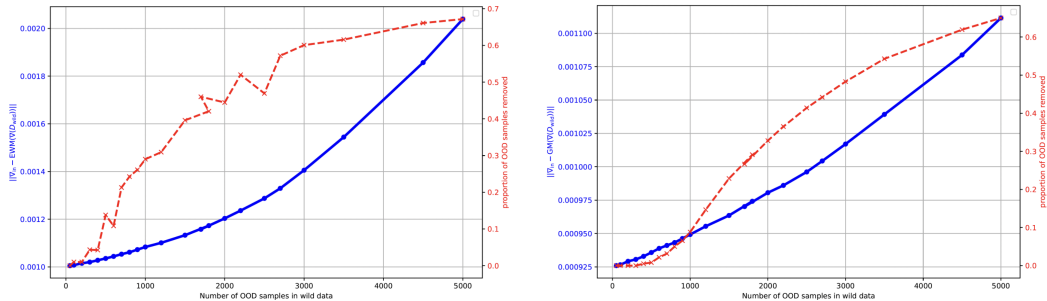


Figure 3: Element-wise median (EWM) identifies more outliers than the geometric median (GM).

Table 4: OOD detection performance of Medix versus recent competitive baselines on CIFAR-100 as InD data. Medix attains the best OOD detection result on every dataset. Performance averaged over five runs; best results in **bold**.

Methods	OOD Datasets									
	SVHN		PLACES365		LSUN-C		TEXTURES		AVERAGE	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
MSP	78.89	79.80	84.38	74.21	83.47	75.28	86.51	72.53	83.31	75.46
Mahalanobis	87.09	80.62	84.63	73.89	84.15	79.43	61.72	84.87	79.40	79.70
ODIN	70.16	84.88	82.16	75.19	76.36	80.10	85.28	75.23	78.49	78.85
Energy	66.91	85.25	81.41	76.37	59.77	86.69	79.01	79.96	71.78	82.07
ReAct	50.93	88.75	83.55	73.10	64.02	80.31	64.40	81.95	65.72	81.03
KNN	46.25	90.39	82.08	75.44	60.85	85.61	62.39	83.95	62.89	83.85
Vim	73.42	84.62	85.34	69.34	86.96	69.74	74.56	76.23	80.07	74.98
VOS	43.24	82.80	76.85	78.63	73.61	84.69	57.57	87.31	62.82	83.36
CSI	44.53	92.65	79.08	76.27	75.58	83.78	61.61	86.47	65.20	84.79
ProxyAnchor	87.21	82.43	70.10	79.84	37.19	91.68	65.64	84.99	65.04	84.74
SSD+	31.19	94.19	77.74	79.90	79.39	85.18	66.63	86.18	63.74	86.36
KNN+	39.23	92.78	80.74	77.58	48.99	89.30	57.15	88.35	56.53	87.00
ASH	52.96	90.19	72.62	76.38	75.18	76.52	56.17	86.75	64.23	82.46
CIDER	23.09	95.16	79.63	73.43	16.16	96.33	43.87	90.42	40.69	88.84
CONJ	46.19	90.44	80.81	75.83	60.45	85.90	62.13	83.77	62.40	83.99
DRL	20.15	94.07	76.64	77.55	16.97	94.63	31.97	92.09	36.43	89.59
Medix	0.48	99.87	24.52	93.42	0.73	99.84	8.99	97.92	8.68	97.76
(Ours)	± 0.05	± 0.02	± 1.76	± 0.36	± 0.08	± 0.03	± 0.72	± 0.24	± 0.65	± 0.16

using stochastic gradient descent with momentum 0.9, and weight decay 10^{-4} for 500 epochs. The initial learning rate is 0.5 with cosine scheduling and the batch size is 512. We use the checkpoints provided by Ming et al. (2023)². The results in Table 4 show that Medix outperforms even these competitive baselines by a significant margin—including those trained with contrastive learning objectives—attaining an average FPR95 of 8.68% and an average AUROC of 97.76%. Beyond raw performance, Medix comes with a provably low error rate, offering theoretical guarantees that most alternative approaches lack.

A.4 Evaluation on Large-Scale, Complex Unseen OOD Settings

We next test whether Medix handles large-scale wild OOD data under the unseen OOD setting. This setting is markedly more challenging for two reasons: (1) it uses a large-scale wild OOD dataset ($\mathbb{P}_{\text{out}}$), raising both data complexity and computational demand, and (2) the test-time OOD distribution ($\mathbb{P}_{\text{out}}^{\text{test}}$) is deliberately chosen to differ from the wild OOD data used during training, i.e., $\mathbb{P}_{\text{out}}^{\text{test}} \neq \mathbb{P}_{\text{out}}$, faithfully reflecting realistic, demanding deployment scenarios. We used CIFAR-100 as the labeled InD data, the 300K Random Images dataset Hendrycks et al. (2019) as the unlabeled wild OOD data, and tested on SVHN as the OOD test data. This setup is especially demanding because the large-scale wild OOD data (300K Random Images) differs sharply from the OOD test data (SVHN). We compare against baselines that also leverage wild data during training. The results in Table 5 make Medix’s superiority plain: it attains a much lower FPR95 (41.29 ± 1.2) and a higher AUROC (87.25 ± 0.6) than every baseline, confirming its effectiveness at distinguishing InD from OOD data even under this severe distribution shift.

Table 5: Medix leads on large-scale unseen OOD data, cutting FPR by nearly half.

Method	FPR ↓	AUROC ↑
OE	68.80 ± 2.8	82.89 ± 1.1
Energy (w/ OE)	69.81 ± 2.4	85.59 ± 1.0
WOODS	69.41 ± 2.7	86.76 ± 0.8
Medix (ours)	41.29 ± 1.2	87.25 ± 0.9

A.5 Evaluating the Impact of Pseudo-Label Quality on OOD Filtering and Robustness

An important question is whether the quality of pseudo-labels $\hat{y}_{\tilde{x}_i}$ —specifically their softmax confidence—affects OOD filtering, and whether a simple pre-filtering step that discards low-confidence pseudo-labels would improve robustness. To answer it, we ran an experiment measuring how filtering low-confidence pseudo-labels affects OOD filtering, using CIFAR-10 as the InD dataset and LSUN-Resize as the OOD dataset. We computed the softmax probabilities for each pseudo-labeled sample

²<https://github.com>

and discarded low-confidence predictions using a threshold of 0.6, which removed 15.98% of the samples from the training set.

The outcome is clear: filtering and no filtering perform essentially identically. Both yield an FPR95 of 0.01%, and filtering nudges AUROC only marginally, from 99.98% to 99.99%. Removing low-confidence pseudo-labels therefore has no meaningful effect on robustness or OOD detection ability, confirming that our method is resilient to noisy or low-confidence labels and that additional filtering steps offer no worthwhile gain.

A.6 Evaluating Computation and Memory Efficiency in Medix Filtering

We now address whether Medix’s filtering phase is computationally feasible and memory-efficient on large datasets. We conducted profiling experiments on an NVIDIA A100-SXM4-80GB GPU using approximately 15,000 unlabeled samples per InD–OOD pair, with CIFAR-10 and CIFAR-100 as the InD datasets and LSUN-Resize as the OOD dataset. As Table 6 shows, GPU memory usage stays modest and the filtering phase completes within a tractable timeframe, even for thousands of unlabeled samples. Medix filtering is thus computationally feasible and imposes no excessive memory overhead, even on large datasets.

Table 6: Profiling results confirm Medix filtering is fast and memory-light on an NVIDIA A100 80GB GPU.

InD–OOD	Wall Clock Time (s)	Peak GPU Memory (MB)	Current GPU Memory (MB)
CIFAR10 – LSUN-Resize	4497.17	99.46	31.74
CIFAR100 – LSUN-Resize	5478.36	99.56	31.79

A.7 Comparison with Semi-Supervised Open-Set Recognition Methods

Our work differs from recent semi-supervised open-set recognition methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) in three fundamental ways. First, in OpenML Saito et al. (2021) the goal is to handle outliers in semi-supervised learning (SSL) while training a standard classifier, whereas our goal is to detect OOD samples from unlabeled wild data and train a dedicated OOD detector. Put simply, SSL methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) aim to improve classification despite outliers, while our approach targets OOD detection directly in an open-world setting where labeled OOD data is unavailable.

Second, these SSL methods train a classifier that is robust to OOD samples, treating outliers as noise to be suppressed; our goal is the opposite—to explicitly detect OOD samples and train an OOD detector. Whereas SSL methods suppress OOD samples during training to boost SSL performance, Medix actively detects and exploits those OOD samples to build a more effective and reliable OOD detection system.

Third, the two approaches use fundamentally different techniques. OpenMatch is based on consistency regularization, enforcing consistency across stochastic transformations of the same input through soft constraints on the classifier’s outputs that encourage smooth decision boundaries. Medix, by contrast, relies on median-based filtering: it harnesses the statistical robustness of the median—its insensitivity to extreme values—to identify potential OOD samples, and our theoretical analysis derives error bounds from the contamination effect (proportion of OOD samples) and the concentration effect (sub-Gaussian behavior of InD gradients), guaranteeing that OOD detection error stays low.

B Broader Impacts and Limitations

As machine learning advances, solving OOD detection is essential for robust and reliable model performance in real-world applications. We introduce a median-centric framework for OOD detection that strengthens OOD handling and model safety. Its impact extends well beyond theory, with direct applications in healthcare, autonomous systems, and finance. By improving OOD detection, we address a key barrier to model deployment and help build greater trust in and adoption of machine

733 learning technologies. Future work could integrate Medix with generative models for outlier synthesis
734 or adapt it to dynamic environments where OOD distributions evolve over time.

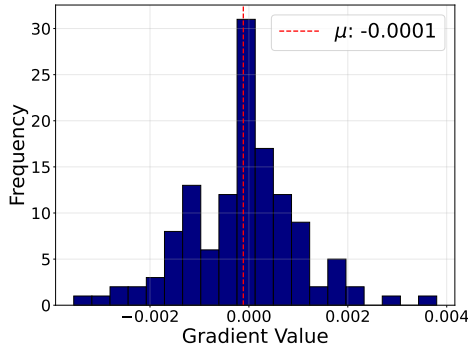
735 One limitation of Medix is its reliance on a mixture of unlabeled InD and OOD data, which in practice
736 may be noisy, corrupted, or inconsistently sampled. This assumption, however, is not specific to our
737 method—it reflects the inherent uncertainty of real-world deployment and is a standard starting point
738 in robust learning theory. We make our assumptions explicit and verifiable by rigorously formalizing
739 them in the main theorems and by stating a precise geometric condition in Remark 4.3 that justifies
740 the separation required for reliable filtering. We further extend our guarantees to looser settings
741 in Theorem C.3, removing the sub-Gaussian tail requirement and establishing the resilience of our
742 framework under weaker distributional assumptions. In short, the assumptions Medix makes are
743 both interpretable and relaxable, providing a principled foundation for future improvements in OOD
744 detection under realistic noise.

745 C Proof of Theorem

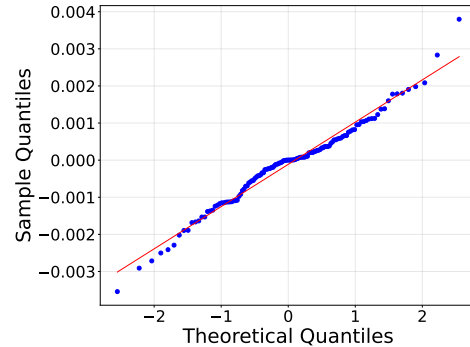
746 For clarity and consistency across the theoretical and empirical sections, Table 7 summarizes the key
 747 notation used throughout the paper, covering gradient-related quantities, sample sizes, filtering terms,
 748 and the parameters appearing in our concentration bounds. Unless stated otherwise, we use boldface
 749 for vectors, $\mathcal{X}$ for the input space, and compute all gradients with respect to a fixed pretrained model
 750 f_ϕ .

Table 7: Notation summary. We use bold symbols for vectors and $\mathcal{X}$ to denote the input space.

Symbol	Meaning
$\mathcal{X}$	Input space (e.g., images in $[0, 1]^d$)
f_ϕ	Model parameterized by ϕ
$\ell(f_\phi(x))$	Loss function evaluated at input x
$\nabla \ell(f_\phi(x))$	Gradient of loss at point x
$\mathbf{g}(x)$	Shorthand for $\nabla \ell(f_\phi(x))$, gradient vector
$\bar{\mathbf{g}}_{\text{in}}$	Empirical mean of gradients over InD points
m	Total number of unlabeled samples in $\mathcal{S}_{\text{wild}}$
$m_{\text{in}} = (1 - \pi)m$	Number of InD samples
$m_{\text{out}} = \pi m$	Number of OOD samples
$\pi \in (0, 1)$	OOD contamination ratio
$\mathcal{S}_{\text{wild}}$	Unlabeled mixture of InD and OOD data
$\mathcal{S}_{\text{in}}^*$	Inliers selected by the median filter
$\mathcal{S}_{\text{out}}^*$	OOD points selected by the median filter
$\mathcal{B}_\varepsilon(\bar{\mathbf{g}}_{\text{in}})$	Ball of radius ε centered at $\bar{\mathbf{g}}_{\text{in}}$
m_ε	Number of InD points inside $\mathcal{B}_\varepsilon$
m_ε^*	Number of selected inliers inside $\mathcal{B}_\varepsilon$
m_{sep}^*	Number of well-separated OODs in selected set
ERR_{in}	Fraction of InD points misclassified as OOD
ERR_{out}	Fraction of OOD points misclassified as InD
σ^2	Sub-Gaussian proxy variance of each gradient coordinate
ε	Threshold for gradient deviation, $\varepsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$
$\delta \in (0, 1)$	Confidence parameter for concentration bounds
d	Dimensionality of gradient vectors



(a) InD gradient histogram.



(b) InD gradient Q-Q plot.

Figure 4: InD sample gradients exhibit sub-Gaussian behavior in each coordinate. (left) Histogram of gradient values (CIFAR-100 InD data) concentrated around the mean with light tails, consistent with sub-Gaussianity. (right) Q-Q plot of empirical InD gradient quantiles against a theoretical Gaussian, confirming alignment with sub-Gaussianity.

751 C.1 Bound on the Inlier Misclassification Rate

752 **Theorem C.1** (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are*
 753 *i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\epsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$, and*
 754 *fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification*
 755 *rate of the element-wise median (EWM) filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

756 *Proof.* We structure our argument into two key steps. In **Step 1**, we establish a high-probability
 757 concentration bound for the gradients of InD samples around their mean, denoted as $\bar{\nabla}_{\text{in}}$. We define
 758 an InD sample whose gradient lies within this high-probability region as a *good inlier*. In **Step**
 759 **2**, we leverage a “swapping” argument on the EWM objective to demonstrate that removing more
 760 than a $(2\eta + \frac{\pi}{2}(1-\pi))$ fraction of *good inliers*—where η represents the fraction of InD samples
 761 with gradients outside the high-probability region (i.e., non-good InD samples)—would contradict
 762 optimality. Combining these two steps, we derive the stated upper bound on the inlier misclassification
 763 rate, ERR_{in} .

764 **Step 1: Concentration of InD Gradients.** Let $\nabla \ell(f_\phi(\mathbf{x})) \in \mathbb{R}^d$ be the gradient (with respect to ϕ)
 765 of a loss evaluated at an InD sample $\mathbf{x}$. Suppose each coordinate of this gradient is sub-Gaussian with
 766 variance proxy σ^2 , centered at the same mean vector $\bar{\nabla}_{\text{in}}$. That is, for each coordinate $j \in \{1, \dots, d\}$,

$$\mathbb{P}\left(|\nabla \ell(f_\phi(\mathbf{x}))_j - \bar{\nabla}_{\text{in},j}| \geq \epsilon\right) \leq 2 \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (6)$$

767 Define

$$\mathbf{v} := \nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}} \in \mathbb{R}^d.$$

768 It is straightforward to show that

$$\|\mathbf{v}\|_2 \leq \sqrt{d} \|\mathbf{v}\|_\infty,$$

769 so if $\|\mathbf{v}\|_2 > \epsilon \sqrt{d}$ then there must exist at least one coordinate j with $|v_j| > \epsilon$.

770 Hence, combining (6) across $j = 1, \dots, d$ via a union bound yields

$$\mathbb{P}\left(\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (7)$$

771 Let $m_{\text{in}} = (1 - \pi)m$ denote the total number of InD points in the set $\mathcal{S}_{\text{wild}}$. We define an InD point
 772 $\mathbf{x}$ to be *bad* if

$$\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}. \quad (8)$$

773 Our goal is to show that, with high probability, only a small fraction of these m_{in} inliers can be bad.

774 From our sub-Gaussian assumption and applying the union bound across the d coordinates, we define
 775 a constant $p \in [0, 1]$ as follows:

$$p := \mathbb{P}\left(\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (9)$$

776 Here, $\mathbf{x}$ is a generic InD sample, and we assume that the samples are i.i.d. across the m_{in} inliers.

777 Let X be the random variable denoting the number of bad inliers among the m_{in} samples. For each
 778 inlier $\mathbf{x}_i$, define the indicator variable

$$I_i := \begin{cases} 1, & \text{if } \|\nabla \ell(f_\phi(\mathbf{x}_i)) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}, \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

779 Then, X can be expressed as

$$X = \sum_{i=1}^{m_{\text{in}}} I_i. \quad (11)$$

780 By the linearity of expectation, we have

$$\mathbb{E}[X] = \sum_{i=1}^{m_{\text{in}}} \mathbb{E}[I_i] \quad (12)$$

$$= \sum_{i=1}^{m_{\text{in}}} \mathbb{P}(\mathbf{x}_i \text{ is bad}) \quad (13)$$

$$\stackrel{\text{from (9)}}{\leq} m_{\text{in}} \cdot p \quad (14)$$

$$\leq m_{\text{in}} \cdot 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (15)$$

781 Observing that X is a sum of m_{in} Bernoulli(p) random variables, each bounded between 0 and 1, we
782 can apply Hoeffding's inequality to obtain, for any $t > 0$,

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (16)$$

783 Setting

$$t := \sqrt{\frac{m_{\text{in}} \log(\frac{1}{\delta})}{2}}, \quad (17)$$

784 we obtain

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (18)$$

785 With probability at least $1 - \delta$, we therefore have

$$X \leq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (19)$$

786 Dividing both sides by m_{in} yields an upper bound on the fraction of bad inliers:

$$\frac{X}{m_{\text{in}}} \leq \frac{\mathbb{E}[X]}{m_{\text{in}}} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (20)$$

$$\stackrel{\text{from (12)}}{\leq} 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (21)$$

$$= \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}. \quad (22)$$

787 Let us denote this fraction by η . By setting $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$ Thus, with probability at least
788 $1 - \delta$, no more than an $O\left(d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{m_{\text{in}}}}\right)$ fraction of inliers have gradients lying
789 outside the radius $\epsilon\sqrt{d}$ around $\bar{\nabla}_{\text{in}}$. These are precisely the *bad inliers*, while the remainder are
790 called *good inliers*.

791 **Step 2: Contradiction via a Swapping Argument** We demonstrate that if $\mathcal{S}^*$ excludes an excessive
 792 number of InD, specifically when

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (23)$$

793 then it is possible to construct a new set $\hat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than that of $\mathcal{S}^*$,
 794 thereby contradicting the optimality of $\mathcal{S}^*$.

795 Let m_ε be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
 796 lies outside the closed ℓ_2 ball

$$B_\varepsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (24)$$

797 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

798 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (25)$$

799 Define the swap size as

$$m'_\varepsilon := \min \{m_\varepsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}}\}. \quad (26)$$

800 This quantity counts how many good inliers (those inside B_ε) are currently excluded from $\mathcal{S}^*$ and
 801 available for swapping in.

802 Construct a new set $\hat{\mathcal{S}}$ by removing m'_ε points from $\mathcal{S}^*$ that lie outside B_ε (OOD points or bad inliers)
 803 and inserting m'_ε good inliers from B_ε in their place. This maintains cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

804 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\varepsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (27)$$

805 which measures the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging, we obtain:

$$m_\varepsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (28)$$

806 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (29)$$

807 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (30)$$

808 We know that at most a η fraction of inliers are outside B_ε , and the fraction of OOD points among all
 809 samples is π . Therefore, the total number of samples in the dataset that lie outside B_ε satisfies:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (31)$$

810 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (32)$$

811 Combining with the previous inequality gives:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (33)$$

812 So $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (34)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (35)$$

813 Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (36)$$

814 so the points lying inside B_ε form a strict majority in $\hat{\mathcal{S}}$.

815 Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ be the j -th coordinate values of all
816 gradients in $\mathcal{S}^*$ and $\hat{X}^{(j)} = \{\hat{x}_1^{(j)}, \dots, \hat{x}_m^{(j)}\}$ the same in $\hat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

817 Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies
818 outside I_j . So for some coordinate j^* , the number of values inside I_{j^*} increases from below $m/2$ to
819 above $m/2$ due to the swaps. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\hat{X}^{(j^*)}) \in I_{j^*}, \quad (37)$$

820 which implies

$$|\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (38)$$

821 For the remaining coordinates $j \neq j^*$, the swapped points either leave the majority unchanged or
822 increase it, so the median distance from μ_j does not increase and possibly decreases.

823 Thus, component-wise:

$$|\text{EWM}_j(G_{\hat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (39)$$

824 with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (40)$$

825 This contradicts the assumption that $\mathcal{S}^*$ is a minimizer of the optimization problem, which concludes
826 the proof.

$$\Rightarrow \quad \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (41)$$

827 □

828 C.2 Bound on the Outlier Retention Rate

829 We now establish a non-asymptotic upper bound on the fraction of OOD points that remain in
830 the subset returned by the MEDIX filter. This result mirrors the inlier misclassification bound
831 (Appendix C.1), but rather than relying on inlier concentration, it exploits a separation assumption
832 between the OOD and InD gradient means.

833 **Theorem C.2** (Outlier Retention Bound under Vector Separation). Assume that for every OOD point
834 $x \sim P_{\text{out}}$, the gradient $g(x) := \nabla \ell(f_{\varphi_{\text{in}}}(x)) \in \mathbb{R}^d$ satisfies:

835 **(I) Sub-Gaussianity.** Each coordinate $g_j(x)$ is sub-Gaussian with variance proxy σ_{out}^2 .

836 **(II) Vector Mean Separation.** The mean OOD gradient vector μ_{out} satisfies

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d}, \quad \text{for some } \Delta > 0.$$

837 Fix any tolerance $\varepsilon \in (0, \Delta)$ and confidence parameter $\delta \in (0, 1)$. Define:

$$p_{\text{out}}(\varepsilon) := 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right), \quad \eta_{\text{out}} := p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}.$$

838 Then with probability at least $1 - \delta$, the outlier retention rate satisfies:

$$\text{ERR}_{\text{out}} := \frac{|\mathcal{S}_{\text{in}}^* \cap \text{OOD}|}{m_{\text{out}}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1 - \pi}{2\pi}. \quad (42)$$

839 *Proof. Step 1: Concentration of OOD Gradients.*

840 By assumption, each coordinate of $g(x)$ is sub-Gaussian, and the mean vector is separated from $\bar{\nabla}_{\text{in}}$
841 by at least $\Delta\sqrt{d}$. Therefore, for any OOD point x , the event

$$\|g(x) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \quad (43)$$

842 has probability at most $p_{\text{out}}(\varepsilon)$.

843 **Counting the “ambiguous” OOD points.** For every OOD example x_i define the indicator

$$J_i := \mathbf{1}\left\{\|g(x_i) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d}\right\}, \quad i = 1, \dots, m_{\text{out}}.$$

844 Each $J_i \sim \text{Bernoulli}(p_{\text{out}}(\varepsilon))$ and the J_i 's are independent. Let

$$Y := \sum_{i=1}^{m_{\text{out}}} J_i, \quad \mu := \mathbb{E}[Y] = m_{\text{out}} p_{\text{out}}(\varepsilon).$$

845 Because the indicators J_i take values in $\{0, 1\}$, Hoeffding's inequality states that for any $s > 0$

$$\Pr\{Y - \mu \geq s\} \leq \exp\left(-\frac{2s^2}{m_{\text{out}}}\right).$$

846 Choose $s = \sqrt{\frac{m_{\text{out}} \log(1/\delta)}{2}}$; then (C.2) equals δ , and with probability at least $1 - \delta$

$$Y \leq m_{\text{out}} \left(p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}\right) = m_{\text{out}} \eta_{\text{out}}.$$

847 Define the high-probability event

$$\mathcal{E}_1 := \{Y \leq m_{\text{out}} \eta_{\text{out}}\}, \quad (35)$$

848 which holds with probability at least $1 - \delta$.

849 **Step 2: Contradiction via a Swapping Argument** Suppose, toward contradiction, that

$$\text{ERR}_{\text{out}} > \eta_{\text{out}} + \frac{1 - \pi}{2\pi}. \quad (44)$$

850 Let $m_{\text{out}}^* := |\mathcal{S}_{\text{in}}^* \cap \text{OOD}|$, and define $m_\varepsilon^* := \#\{\text{ambiguous OODs in } \mathcal{S}_{\text{in}}^*\}$. Then, under $\mathcal{E}_1$,

$$m_{\text{sep}}^* := m_{\text{out}}^* - m_\varepsilon^* > \frac{1-\pi}{2\pi} m_{\text{out}} = \frac{1}{2} m_{\text{in}}. \quad (45)$$

Thus, more than half the points in $\mathcal{S}_{\text{in}}^*$ are separated OODs whose gradients lie outside the ball

$$B_\varepsilon := \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon \sqrt{d} \right\}. \quad (46)$$

By the definition of the coordinate-wise median, this implies that in every coordinate, the majority of entries in $G_{\mathcal{S}_{\text{in}}^*}^{(j)}$ lie outside the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, hence:

$$\|\text{EWM}(G_{\mathcal{S}_{\text{in}}^*}) - \bar{\nabla}_{\text{in}}\|_2^2 \geq \varepsilon^2 d. \quad (47)$$

Now consider a new subset $\hat{\mathcal{S}}$ formed by swapping the m_{sep}^* separated OODs in $\mathcal{S}_{\text{in}}^*$ with an equal number of InD points whose gradients lie inside B_ε . Such inliers exist because fewer than $m_{\text{out}} \eta_{\text{out}}$ are ambiguous, and the total number of InD points is m_{in} .

Then at least half the entries in every coordinate of $G_{\hat{\mathcal{S}}}^{(j)}$ lie inside the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, implying:

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2^2 \leq \varepsilon^2 d. \quad (48)$$

Combining with (47) yields a contradiction:

$$\|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\hat{\mathcal{S}}})\|_2^2 < \|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}_{\text{in}}^*})\|_2^2. \quad (49)$$

This contradicts the optimality of $\mathcal{S}_{\text{in}}^*$, and so the assumption must be false. Therefore, our assumption must be false. With probability at least $1 - \delta$, we conclude that the outlier retention rate is bounded by

$$\text{ERR}_{\text{out}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1-\pi}{2\pi}, \quad (50)$$

□

C.3 Theorem C.3 without Sub-Gaussianity

Theorem C.3 (Inlier Misclassification Bound without Sub-Gaussianity). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and that each coordinate has variance σ^2 and finite fourth moment bounded by μ_4 , i.e.,*

$$\mathbb{E}[(\nabla \ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^2 = \sigma^2, \quad \mathbb{E}[(\nabla \ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^4 \leq \mu_4.$$

Let $m_{\text{in}} = (1 - \pi)m$ denote the number of InD samples in $\mathcal{S}_{\text{wild}}$, and fix any tolerance $\varepsilon > \sigma$ and confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the element-wise median (EWM) filtering rule satisfies:

$$\text{ERR}_{\text{in}} \leq \underbrace{2 \left(\frac{\mu_4 - \sigma^4}{d(\varepsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \right)}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

871 **Differences with the original bound.** The original, sub-Gaussian bound carries a decay term

$$2 \exp\left(-\frac{\epsilon^2 d}{2\sigma_{\text{sub}}^2}\right)$$

872 that decreases exponentially with dimension d , making the bound extremely tight for large d . The
873 new bound, which relies only on finite fourth moments, instead has a decay term

$$\frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}$$

874 that decays as $O(1/d)$, and is therefore looser for large d . The trade-off is deliberate: the original
875 bound requires sub-Gaussian tails, whereas the new one assumes only finite fourth moments and thus
876 admits far heavier tails. Notably, both bounds share the same finite-sample term

$$\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}},$$

877 which shrinks as m_{in} grows, independent of the tail assumptions.

878 *Proof.* The proof follows the structure of the original Theorem 4.1, consisting of two main steps: (1)
879 establishing a concentration bound on the fraction of “bad” InD samples (those far from ∇_{in}) without
880 relying on sub-Gaussianity, and (2) using a swapping argument to bound the inlier misclassification
881 rate. The key difference is the replacement of the sub-Gaussian tail bound with a moment-based
882 bound derived from the finite fourth moment assumption.

883 **Step 1: Concentration of InD Gradients** Define the gradient deviation for an InD sample x as:

$$v = \nabla \ell(f_\phi(x)) - \nabla_{\text{in}} \in \mathbb{R}^d, \quad (51)$$

884 where $\nabla_{\text{in}} = \frac{1}{n} \sum_{(x_i, y_i) \in S_{\text{in}}} \nabla \ell(f_\phi(x_i))$ is the mean gradient over the labeled InD training set S_{in} ,
885 and we assume ∇_{in} is the population mean. An InD sample is classified as “bad” if:

$$\|v\|_2 > \epsilon\sqrt{d}, \quad (52)$$

886 and “good” otherwise. Our goal is to bound the fraction of bad inliers among the m_{in} InD samples in
887 S_{wild} .

888 Since $\|v\|_2^2 = \sum_{j=1}^d v_j^2$, we analyze the ℓ_2 norm via the sum of squared coordinates. Assume that
889 the coordinates $v_j = \nabla \ell(f_\phi(x))_j - \nabla_{\text{in},j}$ are independent across $j = 1, \dots, d$, each with:

$$\mathbb{E}[v_j] = 0 \quad (\text{assuming } \nabla_{\text{in},j} = \mathbb{E}[\nabla \ell(f_\phi(x))_j]), \quad (53)$$

$$\mathbb{E}[v_j^2] = \sigma^2, \quad \mathbb{E}[v_j^4] \leq \mu_4. \quad (54)$$

890 Thus:

$$\mathbb{E}[\|v\|_2^2] = \mathbb{E}\left[\sum_{j=1}^d v_j^2\right] = d\sigma^2, \quad (55)$$

$$\text{Var}(\|v\|_2^2) = \text{Var}\left(\sum_{j=1}^d v_j^2\right) = \sum_{j=1}^d \text{Var}(v_j^2). \quad (56)$$

891 Compute the variance of v_j^2 :

$$\text{Var}(v_j^2) = \mathbb{E}[v_j^4] - (\mathbb{E}[v_j^2])^2 \leq \mu_4 - \sigma^4, \quad (57)$$

892 so:

$$\text{Var}(\|v\|_2^2) \leq d(\mu_4 - \sigma^4). \quad (58)$$

893 Using Chebyshev's inequality for $\|v\|_2^2$:

$$\mathbb{P}(|\|v\|_2^2 - d\sigma^2| \geq t) \leq \frac{d(\mu_4 - \sigma^4)}{t^2}. \quad (59)$$

894 We need $\mathbb{P}(\|v\|_2 > \epsilon\sqrt{d})$, which corresponds to $\|v\|_2^2 > \epsilon^2 d$. Set the threshold:

$$\|v\|_2^2 > \epsilon^2 d \implies \|v\|_2^2 - d\sigma^2 > \epsilon^2 d - d\sigma^2 = d(\epsilon^2 - \sigma^2). \quad (60)$$

895 Thus:

$$\mathbb{P}(\|v\|_2^2 > \epsilon^2 d) = \mathbb{P}(\|v\|_2^2 - d\sigma^2 > d(\epsilon^2 - \sigma^2)) \leq \frac{d(\mu_4 - \sigma^4)}{[d(\epsilon^2 - \sigma^2)]^2} = \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}, \quad (61)$$

896 provided $\epsilon > \sigma$ so that $\epsilon^2 - \sigma^2 > 0$. Define:

$$p = \mathbb{P}(\|\nabla \ell(f_\phi(x)) - \nabla_{\text{in}}\|_2 > \epsilon\sqrt{d}) \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (62)$$

897 Let X be the number of bad inliers among the m_{in} InD samples in S_{wild} , where each InD sample is
898 bad with probability at most p . Then:

$$\mathbb{E}[X] \leq m_{\text{in}}p \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (63)$$

899 Since $X = \sum_{i=1}^{m_{\text{in}}} I_i$, where $I_i = 1$ if the i -th InD sample is bad and 0 otherwise, and I_i are i.i.d.
900 Bernoulli(p) variables bounded in $[0, 1]$, apply Hoeffding's inequality:

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (64)$$

901 Set $t = \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}$, so:

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (65)$$

902 With probability at least $1 - \delta$:

$$X \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (66)$$

903 The fraction of bad inliers is:

$$\eta = \frac{X}{m_{\text{in}}} \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}. \quad (67)$$

904 This η represents the upper bound on the fraction of InD samples whose gradients deviate from $\bar{\nabla}_{\text{in}}$
 905 by more than $\epsilon\sqrt{d}$, relying only on finite fourth moments and coordinate independence.

906 **Step 2: Contradiction via a Swapping Argument** We demonstrate that if $\mathcal{S}^*$ excludes an excessive
 907 number of InD, specifically when

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (68)$$

908 then it is possible to construct a new set $\hat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than that of $\mathcal{S}^*$,
 909 thereby contradicting the optimality of $\mathcal{S}^*$.

910 Let m_ϵ be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
 911 lies outside the closed ℓ_2 ball

$$B_\epsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \epsilon\sqrt{d} \right\}. \quad (69)$$

912 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

913 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (70)$$

914 Define the swap size as

$$m'_\epsilon := \min \{ m_\epsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}} \}. \quad (71)$$

915 This quantity counts how many good inliers (those inside B_ϵ) are currently excluded from $\mathcal{S}^*$ and
 916 available for swapping in.

917 Construct a new set $\hat{\mathcal{S}}$ by removing m'_ϵ points from $\mathcal{S}^*$ that lie outside B_ϵ (OOD points or bad inliers)
 918 and inserting m'_ϵ good inliers from B_ϵ in their place. This maintains cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

919 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\epsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (72)$$

920 which measures the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging, we obtain:

$$m_\epsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (73)$$

921 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (74)$$

922 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (75)$$

923 We know that at most a η fraction of inliers are outside B_ε , and the fraction of OOD points among all
 924 samples is π . Therefore, the total number of samples in the dataset that lie outside B_ε satisfies:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (76)$$

925 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (77)$$

926 Combining with the previous inequality gives:

$$(\text{ERR}_{\text{in}} - \eta) m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (78)$$

927 So $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (79)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (80)$$

928 Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (81)$$

929 so the points lying inside B_ε form a strict majority in $\hat{\mathcal{S}}$.

930 Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ be the j -th coordinate values of all
 931 gradients in $\mathcal{S}^*$ and $\hat{X}^{(j)} = \{\hat{x}_1^{(j)}, \dots, \hat{x}_m^{(j)}\}$ the same in $\hat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

932 Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies
 933 outside I_j . So for some coordinate j^* , the number of values inside I_{j^*} increases from below $m/2$ to
 934 above $m/2$ due to the swaps. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\hat{X}^{(j^*)}) \in I_{j^*}, \quad (82)$$

935 which implies

$$|\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (83)$$

936 For the remaining coordinates $j \neq j^*$, the swapped points either leave the majority unchanged or
 937 increase it, so the median distance from μ_j does not increase and possibly decreases.

938 Thus, component-wise:

$$|\text{EWM}_j(G_{\hat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (84)$$

939 with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (85)$$

940 This contradicts the assumption that $\mathcal{S}^*$ is a minimizer of the optimization problem, which concludes
 941 the proof.

$$\Rightarrow \quad \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (86)$$

942

□